ROCSCIENCE

Case3_with_surcharge
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

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Slide2 Analysis Information

Case3_with_surcharge

Project Summary

File Name:	Case3_with_surcharge.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:03.280s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Loading

1 Distributed Load present

Distributed Load 1	
Distribution:	Constant
Magnitude [kPa]:	100
Orientation:	Vertical

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	3 kPa
Phi	19.6 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	0.609948
Center:	49.898, 35.068
Radius:	0.189
Left Slip Surface Endpoint:	49.806, 34.903
Right Slip Surface Endpoint:	50.074, 35.000
Resisting Moment:	0.637551 kN-m
Driving Moment:	1.04526 kN-m
Total Slice Area:	0.0164126 m2
Surface Horizontal Width:	0.267979 m
Surface Average Height:	0.061246 m

Method: spencer

FS	0.624131
Center:	49.898, 35.068
Radius:	0.189
Left Slip Surface Endpoint:	49.806, 34.903
Right Slip Surface Endpoint:	50.074, 35.000
Resisting Moment:	0.652376 kN-m
Driving Moment:	1.04526 kN-m
Resisting Horizontal Force:	2.55339 kN
Driving Horizontal Force:	4.09112 kN
Total Slice Area:	0.0164126 m2
Surface Horizontal Width:	0.267979 m
Surface Average Height:	0.061246 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	12060
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	11460
Number of Invalid Surfaces:	600

Error Codes

Error Code -108 reported for 373 surfaces
Error Code -111 reported for 112 surfaces
Error Code -112 reported for 115 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge
-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.609948

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.0107192	0.00116804	-27.3194	Material 1	3	19.6	7.1332	4.35088	3.79371	0	3.79371	0.108929	0.108929
2	0.0107192	0.00341517	-23.7088	Material 1	3	19.6	6.86426	4.18684	3.33303	0	3.33303	0.318576	0.318576
3	0.0107192	0.00549143	-20.1959	Material 1	3	19.6	6.64437	4.05272	2.95639	0	2.95639	0.512281	0.512281
4	0.0107192	0.00740914	-16.7609	Material 1	3	19.6	6.45731	3.93862	2.63596	0	2.63596	0.691189	0.691189
5	0.0107192	0.00917765	-13.3871	Material 1	3	19.6	6.29257	3.83814	2.35378	0	2.35378	0.856179	0.856179
6	0.0107192	0.010804	-10.0601	Material 1	3	19.6	6.1431	3.74697	2.09774	0	2.09774	1.0079	1.0079
7	0.0107192	0.0122931	-6.76717	Material 1	3	19.6	6.00388	3.66205	1.85926	0	1.85926	1.14683	1.14683
8	0.0107192	0.0136487	-3.49663	Material 1	3	19.6	5.87123	3.58115	1.63205	0	1.63205	1.27329	1.27329
9	0.0107192	0.0148727	-0.237502	Material 1	3	19.6	5.74235	3.50254	1.41129	0	1.41129	1.38749	1.38749
10	0.0107192	0.0159658	3.02085	Material 1	3	19.6	5.615	3.42486	1.19314	0	1.19314	1.48946	1.48946
11	0.0107192	0.0169276	6.28905	Material 1	3	19.6	5.48733	3.34699	0.974447	0	0.974447	1.57919	1.57919
12	0.0107192	0.017756	9.57801	Material 1	3	19.6	5.35771	3.26792	0.752412	0	0.752412	1.65649	1.65649
13	0.0107192	0.018448	12.8993	Material 1	3	19.6	5.22466	3.18677	0.524508	0	0.524508	1.72104	1.72104
14	0.0107192	0.0189986	16.2654	Material 1	3	19.6	5.08675	3.10265	0.288279	0	0.288279	1.77241	1.77241
15	0.0107192	0.0194012	19.6905	Material 1	3	19.6	4.94252	3.01468	0.0412198	0	0.0412198	1.80997	1.80997
16	0.0107192	0.0196468	23.1909	Material 1	3	19.6	4.79038	2.92188	-0.219373	0	-0.219373	1.83289	1.83289
17	0.0107192	0.0197235	26.7859	Material 1	3	19.6	4.62856	2.82318	-0.496574	0	-0.496574	1.84005	1.84005
18	0.0107192	0.0196157	30.4992	Material 1	3	19.6	4.45488	2.71725	-0.794059	0	-0.794059	1.82999	1.82999
19	0.0107192	0.0188475	34.3611	Material 1	3	19.6	41.374	25.236	62.4459	0	62.4459	90.7341	90.7341
20	0.0107192	0.0171578	38.4113	Material 1	3	19.6	43.9077	26.7814	66.7861	0	66.7861	101.601	101.601
21	0.0107192	0.0151863	42.7049	Material 1	3	19.6	41.6721	25.4178	62.9566	0	62.9566	101.417	101.417
22	0.0107192	0.0128797	47.3227	Material 1	3	19.6	39.1878	23.9025	58.7009	0	58.7009	101.202	101.202
23	0.0107192	0.0101419	52.3932	Material 1	3	19.6	36.3223	22.1547	53.7927	0	53.7927	100.947	100.947
24	0.0107192	0.00680082	58.1478	Material 1	3	19.6	32.8248	20.0214	47.8017	0	47.8017	100.635	100.635
25	0.0107192	0.00247572	65.1035	Material 1	3	19.6	28.094	17.1359	39.6981	0	39.6981	100.231	100.231

Global Minimum Query (spencer) - Safety Factor: 0.624131

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.0107192	0.00116804	-27.3194	Material 1	3	19.6	14.7059	9.17839	17.351	0	17.351	9.75439	9.75439
2	0.0107192	0.00341517	-23.7088	Material 1	3	19.6	12.2454	7.64272	13.0383	0	13.0383	7.66067	7.66067
3	0.0107192	0.00549143	-20.1959	Material 1	3	19.6	10.7651	6.71884	10.4437	0	10.4437	6.48383	6.48383
4	0.0107192	0.00740914	-16.7609	Material 1	3	19.6	9.77	6.09776	8.69952	0	8.69952	5.75706	5.75706
5	0.0107192	0.00917765	-13.3871	Material 1	3	19.6	9.0461	5.64595	7.4307	0	7.4307	5.27777	5.27777
6	0.0107192	0.010804	-10.0601	Material 1	3	19.6	8.48746	5.29729	6.45155	0	6.45155	4.9458	4.9458
7	0.0107192	0.0122931	-6.76717	Material 1	3	19.6	8.03597	5.0155	5.66019	0	5.66019	4.70663	4.70663
8	0.0107192	0.0136487	-3.49663	Material 1	3	19.6	7.65716	4.77907	4.99622	0	4.99622	4.52834	4.52834
9	0.0107192	0.0148727	-0.237502	Material 1	3	19.6	7.32925	4.57441	4.42147	0	4.42147	4.39109	4.39109
10	0.0107192	0.0159658	3.02085	Material 1	3	19.6	7.0378	4.39251	3.91061	0	3.91061	4.28201	4.28201
11	0.0107192	0.0169276	6.28905	Material 1	3	19.6	6.77276	4.22709	3.44607	0	3.44607	4.19248	4.19248
12	0.0107192	0.017756	9.57801	Material 1	3	19.6	6.52692	4.07365	3.01516	0	3.01516	4.11653	4.11653
13	0.0107192	0.018448	12.8993	Material 1	3	19.6	6.29483	3.9288	2.60837	0	2.60837	4.04999	4.04999
14	0.0107192	0.0189986	16.2654	Material 1	3	19.6	6.0723	3.78991	2.21832	0	2.21832	3.99001	3.99001
15	0.0107192	0.0194012	19.6905	Material 1	3	19.6	5.8559	3.65485	1.83904	0	1.83904	3.93466	3.93466
16	0.0107192	0.0196468	23.1909	Material 1	3	19.6	5.64274	3.52181	1.46542	0	1.46542	3.88284	3.88284
17	0.0107192	0.0197235	26.7859	Material 1	3	19.6	5.43022	3.38917	1.09291	0	1.09291	3.83423	3.83423
18	0.0107192	0.0196157	30.4992	Material 1	3	19.6	5.21583	3.25536	0.717137	0	0.717137	3.7894	3.7894
19	0.0107192	0.0188475	34.3611	Material 1	3	19.6	40.3226	25.1666	62.251	0	62.251	89.8203	89.8203
20	0.0107192	0.0171578	38.4113	Material 1	3	19.6	41.6852	26.017	64.6393	0	64.6393	97.692	97.692
21	0.0107192	0.0151863	42.7049	Material 1	3	19.6	38.5983	24.0904	59.2288	0	59.2288	94.8524	94.8524
22	0.0107192	0.0128797	47.3227	Material 1	3	19.6	35.3647	22.0722	53.5609	0	53.5609	91.9157	91.9157
23	0.0107192	0.0101419	52.3932	Material 1	3	19.6	31.8686	19.8902	47.4332	0	47.4332	88.8053	88.8053
24	0.0107192	0.00680082	58.1478	Material 1	3	19.6	27.9068	17.4175	40.4891	0	40.4891	85.4067	85.4067
25	0.0107192	0.00247572	65.1035	Material 1	3	19.6	23.0258	14.3711	31.9337	0	31.9337	81.5464	81.5464

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.609948

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	49.8059	34.9029	0	0	0
2	49.8166	34.8974	0.0974676	0	0
3	49.8273	34.8927	0.186736	0	0
4	49.838	34.8887	0.269614	0	0
5	49.8487	34.8855	0.34734	0	0
6	49.8595	34.883	0.420795	0	0
7	49.8702	34.8811	0.490633	0	0
8	49.8809	34.8798	0.557354	0	0
9	49.8916	34.8791	0.621357	0	0
10	49.9023	34.8791	0.682972	0	0
11	49.9131	34.8797	0.742485	0	0
12	49.9238	34.8808	0.800152	0	0
13	49.9345	34.8827	0.856221	0	0
14	49.9452	34.8851	0.910937	0	0
15	49.9559	34.8882	0.96456	0	0
16	49.9667	34.8921	1.01738	0	0
17	49.9774	34.8967	1.06974	0	0
18	49.9881	34.9021	1.12204	0	0
19	49.9988	34.9084	1.1748	0	0
20	50.0095	34.9157	1.16064	0	0
21	50.0203	34.9242	1.06365	0	0
22	50.031	34.9341	0.8875	0	0
23	50.0417	34.9457	0.625128	0	0
24	50.0524	34.9597	0.265905	0	0
25	50.0631	34.9769	-0.206972	0	0
26	50.0739	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 0.624131

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	49.8059	34.9029	0	0	0
2	49.8166	34.8974	0.253699	0.103695	22.2314
3	49.8273	34.8927	0.446326	0.182427	22.2313
4	49.838	34.8887	0.602891	0.24642	22.2313
5	49.8487	34.8855	0.735695	0.300701	22.2313
6	49.8595	34.883	0.851612	0.34808	22.2313
7	49.8702	34.8811	0.954853	0.390278	22.2314
8	49.8809	34.8798	1.04819	0.428426	22.2313
9	49.8916	34.8791	1.13353	0.463309	22.2314
10	49.9023	34.8791	1.21229	0.495498	22.2313
11	49.9131	34.8797	1.28551	0.525427	22.2313
12	49.9238	34.8808	1.35403	0.553434	22.2314
13	49.9345	34.8827	1.41853	0.579799	22.2314
14	49.9452	34.8851	1.4796	0.604759	22.2314
15	49.9559	34.8882	1.53775	0.628526	22.2314
16	49.9667	34.8921	1.59346	0.651297	22.2314
17	49.9774	34.8967	1.64721	0.673267	22.2314
18	49.9881	34.9021	1.6995	0.694639	22.2314
19	49.9988	34.9084	1.75088	0.715639	22.2314
20	50.0095	34.9157	1.72684	0.705815	22.2314
21	50.0203	34.9242	1.62425	0.663882	22.2314
22	50.031	34.9341	1.45201	0.593482	22.2314
23	50.0417	34.9457	1.20839	0.493908	22.2314
24	50.0524	34.9597	0.889909	0.363733	22.2313
25	50.0631	34.9769	0.490464	0.200468	22.2314
26	50.0739	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	70, 20 70, 35 50, 35 30, 25 20, 25 20, 20

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Distributed Load	50, 35 70, 35	Constant DistributionOrientation: VerticalMagnitude: 100 kN/m2Creates Excess Pore Pressure: No